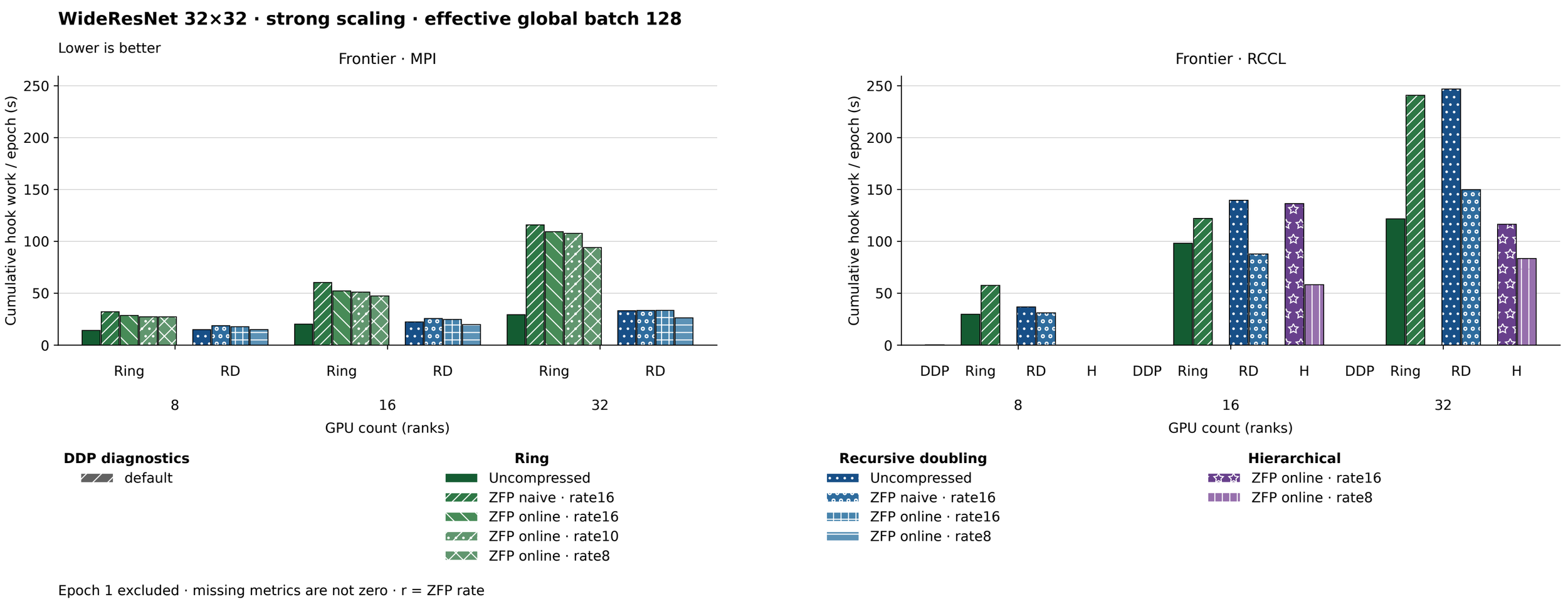

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI
Frontier · RCCL
Cumulative hook work / epoch (s)
0
50
100
150
200
250
Ring
RD
DDP
H
8
16
32
GPU count (ranks)
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate